VIT UNIVERSITY

(School of Computer Science Engineering)

PROJECT REPORT ON

“AI-Based Job Portal with Skill Trend Analysis”

Submitted by:

Name: [ADARSH MISHRA]
Registration Number: [25BCE10947]
PROGRAM: B.TECH
Semester: [2ND]

Under the Guidance of:

Faculty Name: [M K JAYANTHI]

Title: AI-Based Job Portal with Skill Trend Analysis

---

1. Introduction

In today's time, students and job seekers often struggle to understand which skills are in demand in the market. At the same time, recruiters also find it difficult to quickly filter suitable candidates based on skills and experience.

This project is a simple job portal system that allows applicants to apply for jobs and recruiters to search for suitable candidates. Additionally, the system tracks skill demand and shows trending skills.

---

2. Problem Statement

There is no simple system for:

- Managing applicant data based on skills and experience
- Helping recruiters find suitable candidates easily
- Understanding which job skills are currently in demand

This project aims to solve these problems using a database-driven approach with basic data analysis.

---

3. Objective

The main objectives of this project are:

- To create a system where applicants can apply for jobs
- To allow recruiters to filter candidates based on requirements
- To track and analyze skill demand

- To display the most trending job skills

---

4. Methodology

The project is implemented using Python and a database.

- Programming Language: Python
- Database: SQLite
- Approach:
  - Store applicant details in a table
  - Store skills and their search count in another table
  - Update search count whenever a recruiter searches for a skill
  - Display top skills based on search frequency

---

5. System Design

Tables Used:

1. Applicant Table

- Name
- Years of Experience
- Skill

2. Skills Table

- Skill Name
- Number of Searches

---

6. Features

- Apply for job (store applicant details)
- Recruit applicant (filter by skill + experience)
- Skill demand tracking
- Trending skills display (Top 5)

---

7. Implementation

The system is menu-driven and has three main options:

1. Apply for Job
2. Recruit Applicant
3. Current Demanding Skills

When a recruiter searches for a skill, its search count is increased, which helps in identifying trends.

---

8. Challenges Faced

- Understanding how to connect Python with database
- Writing correct SQL queries
- Managing user input properly
- Avoiding duplicate data entries

---

9. Results

The system successfully:

- Stores applicant data
- Filters candidates based on requirements
- Tracks skill demand

- Displays trending skills

---

10. Future Improvements

- Add machine learning to predict future skill trends
- Create a graphical user interface (GUI)
- Add login system for users
- Integrate real job datasets

---

11. Conclusion

This project helped in understanding how databases and basic data analysis can be used to solve real-world problems. It also gave practical experience in Python and SQL.